

Note that trying to move the switching point voltages of each inverter to differing levels makes the problem of metastability worse. When each inverter's V_{sp} is the same the gain around the loop is the highest (and so it's possible noise alone can cause the inverters to snap to a valid logic state).

Figure 13.18 shows a level-sensitive latch. When the clock signal is high, the data or D input passes through the NMOS PG and drives the cross-coupled inverters. The feedback inverter's lengths are made long to ensure that the source driving the D input can "overpower" the feedback inverter and allow the output to switch states. This circuit, while simple, has several fundamental issues. To begin, we know that an NMOS pass gate's output will swing from 0 to $V_{DD} - V_{THN}$. To optimize the noise margins, we lower the V_{sp} of I1. Here we reduce the width of the PMOS from 20 to 10. Another concern is the contention current that flows when the D input is fighting with the output of I2 for control of I1's input. By increasing the lengths in I2 to 10, we lower this wasted current.

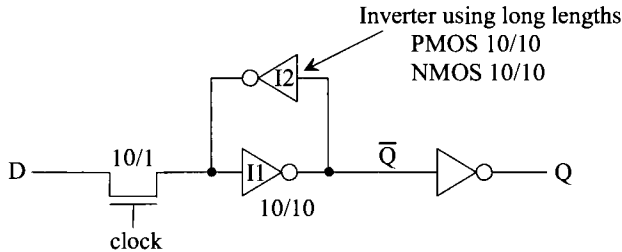


Figure 13.18 A level-sensitive latch.

Figure 13.19 shows the input and output simulation results for the latch in Fig. 13.18. When the clock signal is high, the input flows to the output (with a delay). When the clock signal goes low, the value on the D input is captured and remains on the Q output until the clock signal goes back high again. Notice that at 8 ns, when the clock signal goes high, the delay between the D input going high and the Q output going high is considerable.

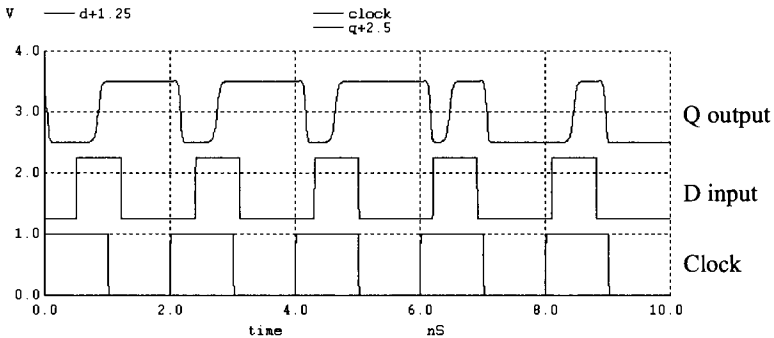


Figure 13.19 Simulating the level-sensitive latch in Fig. 13.18.

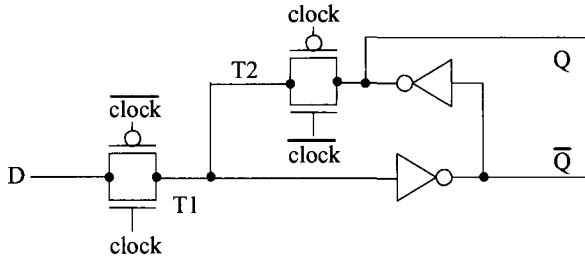


Figure 13.20 A higher performance level-sensitive latch.

Figure 13.20 shows a modification of the latch in Fig. 13.18. A TG is used on the input of the latch instead of a PG. This improves the noise performance of the circuit (at the cost of an additional clock signal, that is, the complement of clock). We've also added a TG in series with the feedback inverter. When T1 is on, the Q output follows the D input. When T1 is on, T2 is off so that the input doesn't have to fight with the feedback inverter. When T1 turns off, T2 turns on and the value of D at that instance is stored in the inverters. Figure 13.21 shows the simulation results using this latch with the input signals used to generate Fig. 13.19. Notice, for example, the improvement in performance over what's seen in Fig. 13.19 when the clock transitions high at 8 ns.

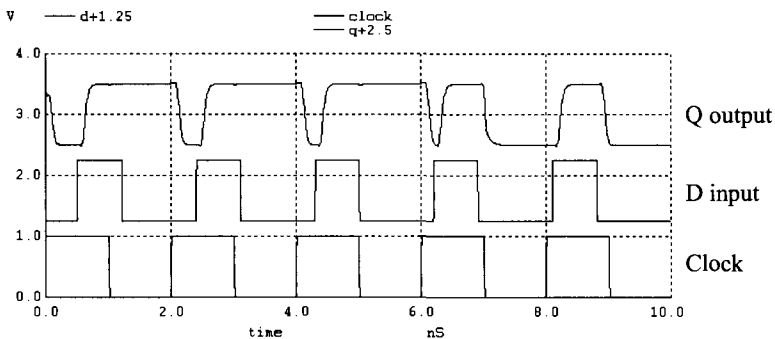


Figure 13.21 Simulating the level-sensitive latch in Fig. 13.20.

An Edge-Triggered D-FF

Notice that the value of the D input in the circuit seen in Fig. 13.20 is stored or captured when clock goes low. To implement an edge-triggered FF, we can use two of these level-sensitive latches in cascade. When the clock is low, the first stage tracks the D input and the second stage holds the previous output. When clock goes high, the first stage captures the input and transfers it to the second stage. The first stage is often called the “master” latch, while the second stage is the “slave” latch. Figure 13.22 shows an implementation of an edge-triggered D-FF. When clock is low, T1 and T4 are on. T2 and T3 are off. The D input flows through to point A and its complement to point B. When clock goes high, T1 and T4 shut off while T2 and T3 turn on. This causes the value of the D input, when

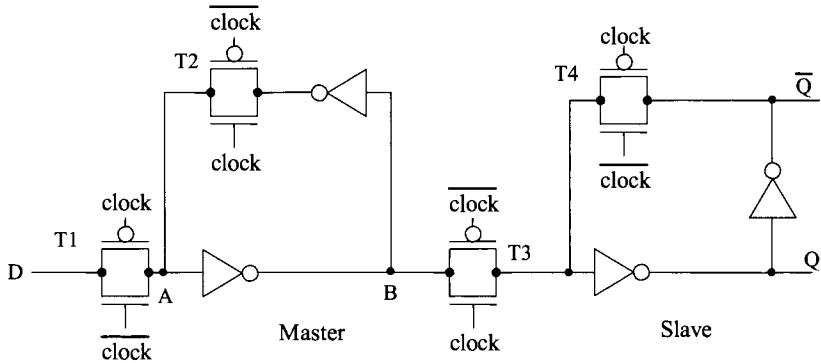


Figure 13.22 An edge-triggered D-FF.

clock transitioned high, to be captured and passed to the Q output. When clock goes back low, T1 and T4 turn on. The value of D on the Q output is then circulated around the two inverters. The Q output can change states again when clock goes back high. Figure 13.23 shows results of simulating this FF with SPICE. Note the output only changes on the rising edge of the clock signal.

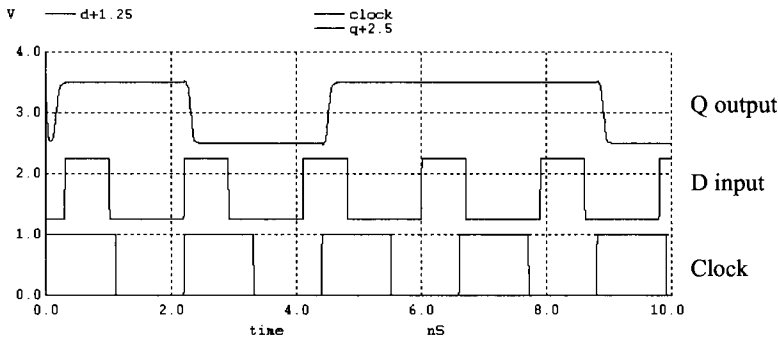


Figure 13.23 Simulating the operation of the edge-triggered FF in Fig. 13.22.

Figure 13.24 shows the addition of NAND gates to the circuit of Fig. 13.22 for clear and set inputs. When clear goes low, the outputs of N1 and N4 go high. If T3 is on or T4 is on, the outputs of the FF are forced to $Q = 0$ (and thus its complement is set to a 1). If the set input goes low, then Q is forced to a 1. If both clear and set are pulled low, then both the Q output and its complement are driven high. This feature can be useful in some situations.

Note that it is **very important** for the clock signals we use to have quick rise times. If they don't, then the FF may not function correctly. Often the clock signals are applied to a string of inverters both to sharpen their transition times and to generate the complement clock required for the TGs.

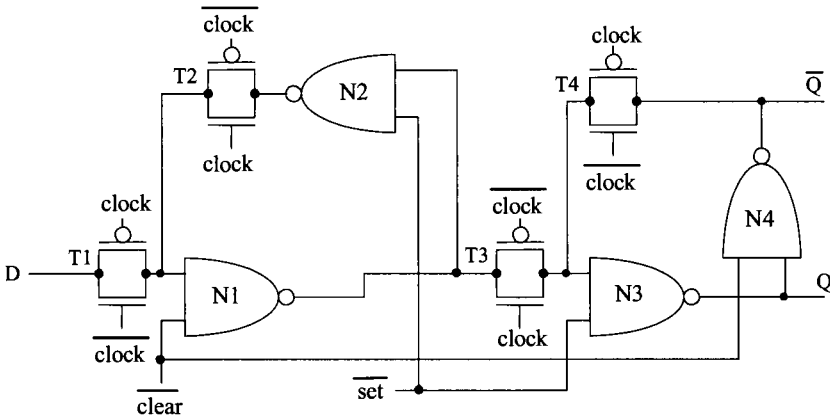


Figure 13.24 An edge-triggered FF with asynchronous set and clear.

Flip-Flop Timing

The data must be *set up* or present on the D input of the FF (see Fig. 13.22) a certain time before we apply the clock signal. This time is defined as the setup time of the FF. To understand the origin of this time, consider the time it takes the signal at D to propagate through T1 and the inverter to node B. Before the clock pulse can be applied, the logic level D must be settled on node B. Consider the waveforms of Fig. 13.25. The time between D going high (or low) and the clock rising edge is termed the setup time of the FF and is labeled, t_s .

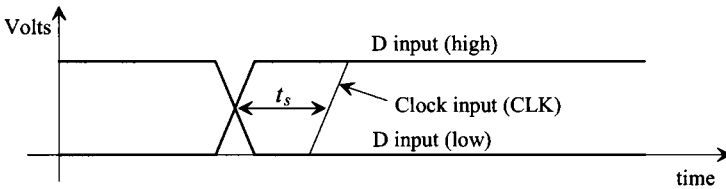


Figure 13.25 Illustrating D FF setup time.

The wanted D input must be applied t_s before the clock pulse is applied. Now the question becomes “How long does the wanted D input have to remain applied to, or *held* on, the input of the FF after the clock pulse is applied?” This time is called the hold time, t_h , and is illustrated in Fig. 13.26. Shown in this figure, t_h is a positive number. However, inspection of Fig. 13.22 shows that if the D input is removed slightly before the clock pulse is applied, node B will remain unchanged because of the propagation delay from the D input to node B. Analysis of this FF would yield a negative hold time. In other words, for the point B to charge to D, a time labeled t_s is needed. Once point B is charged, the D input can be changed as long as the clock signal occurs within t_h .

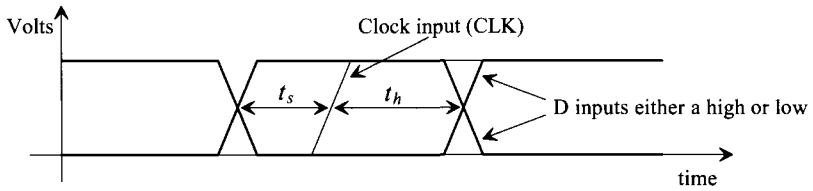


Figure 13.26 Illustrating D FF hold time.

One final important comment regarding the clock input of an FF is (again) in order. If the clock input risetime is slow, the FF will not function properly. There will not be an abrupt transition between the sets of transmission gates turning on and off. The result will be logic levels at indeterminate states. What is usually done to eliminate this problem is to buffer the clock input through several inverters. This has the effect of speeding up the leading and trailing edges of a slow input pulse and presenting a lower input capacitance on the clock input to whatever is driving the FF. The main disadvantage is the increase in delay times, t_{PHL} and t_{PLH} (defined by clock to output), of the FF. In general, the FF of Fig 13.22 should not be laid out without buffering the clock inputs.

The minimum pulse width of the clock, set, or clear inputs in Figs. 13.22 and 13.24 is labeled t_w . The minimum width is determined by the delay through (referring to Fig. 13.20) two NAND gates and a TG. The last timing definition we will comment on here is the recovery time, that is, the time between removing the set or clear inputs and a valid clock input. This variable is labeled t_{rec} .

13.4 Examples

In this section we give examples of delay calculations and comments on how to decrease the delays and the associated performance costs. Throughout this section we use the 50 nm short-channel process.

Example 13.3

Estimate the input capacitance and the delay through the circuit seen in Fig. 13.27. Compare the hand calculation estimates to simulation results.

Using the information seen in Table 10.2, we can calculate the first inverter's input capacitance, and thus the input capacitance of the circuit,

$$C_{in1} = \frac{3}{2} \cdot (C_{oxn1} + C_{oxp1}) = 0.938 \text{ fF} + 1.875 \text{ fF} = 2.81 \text{ fF}$$

Since the widths of the second inverter are 10 times larger than the widths of the first inverter, the input capacitance of the second inverter is 28.1 fF.

The delay through the first inverter, that is, the delay from the node labeled "In" to the node labeled "N1" in Fig. 13.27 is calculated as

$$t_{PHL1} + t_{PLH1} = 0.7 \cdot (3.4k + 3.4k) \cdot (0.625 + 1.25 + 9.38 + 18.75) \text{ fF} = 143 \text{ ps}$$

noting, because $t_{PHL} = t_{PLH}$, that $t_{PHL1} = t_{PLH1} = 71.5 \text{ ps}$.

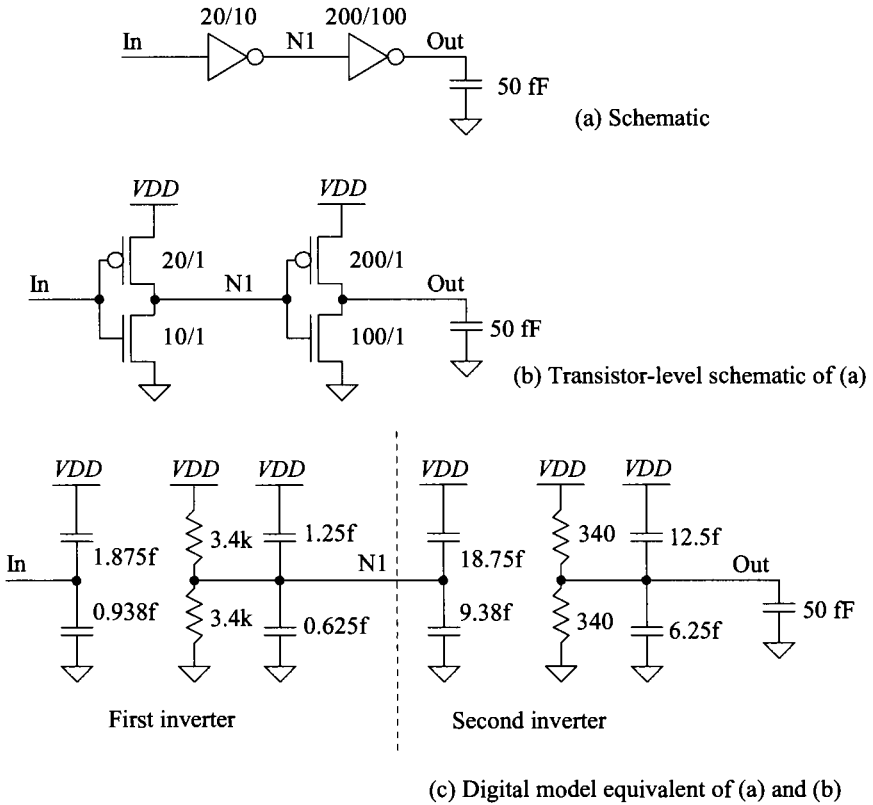


Figure 13.27 Circuit used in Ex. 13.3.

The delay through the second stage, that is, from N1 to the output is calculated as

$$t_{PHL2} + t_{PLH2} = 0.7 \cdot (340 + 340) \cdot 68.75 \text{ fF} = 33 \text{ ps}$$

Again as seen in the first inverter, the effective switching resistance of the PMOS is equal to the NMOS device's effective resistance so $t_{PHL2} = t_{PLH2} = 16.5 \text{ ps}$.

The overall delay through both inverters is

$$t_{PHL} = t_{PLH} = 88 \text{ ps}$$

Figure 13.28 shows the SPICE simulation results. The delays, from the simulations, are approximately 120 ps.

To decrease the delay through the circuit, the simplest solution is to increase the widths of the first inverter. Of course, this increases the circuit's input capacitance. Note that the *analysis* of this circuit has nothing to do with the *design* equations used for implementing a buffer, Sec. 11.4. ■

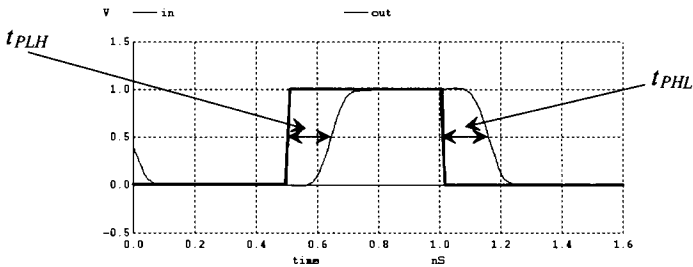


Figure 13.28 Simulating the circuit in Ex. 13.3 and Fig. 13.27.

Example 13.4

In Ex. 13.3 the devices were sized to give (ideally) equal propagation delays. Repeat Ex. 13.3 for the circuit seen in Fig. 13.29.

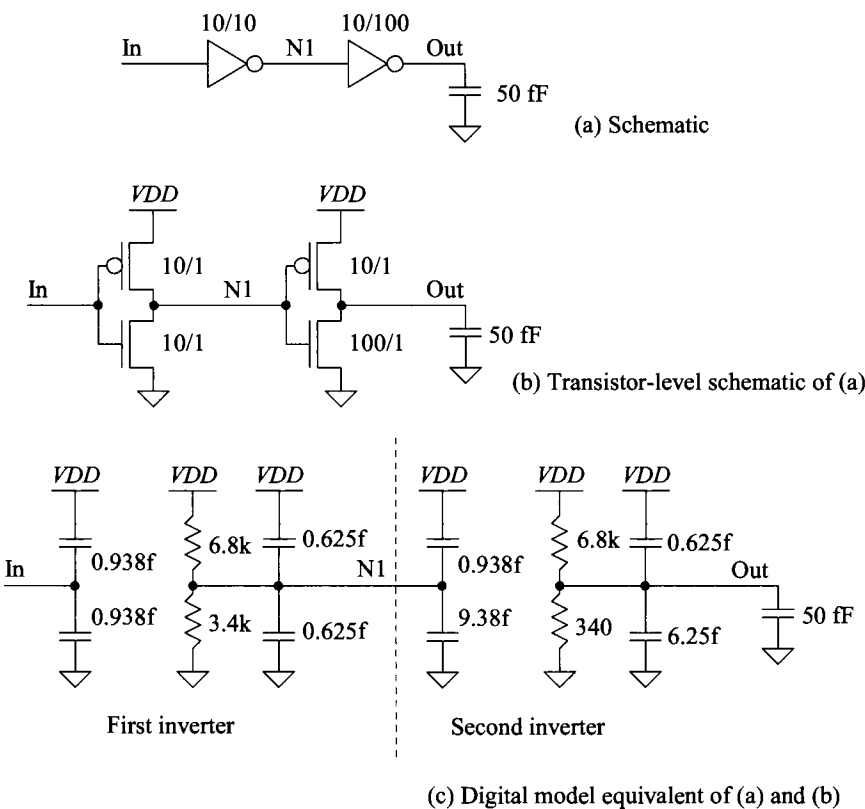


Figure 13.29 Circuit used in Ex. 13.4.

The PMOS device has a width of 10 which is half of the width used for the data in Table 10.2. We can write the effective switching resistance and oxide capacitance for this device as

$$R_p = 6.8k \text{ and } C_{oxp} = 0.625 \text{ fF}$$

As seen in Fig. 13.29, the input capacitance of the circuit is 1.876 fF.

Calculating t_{PLH} begins by noting that when the input goes high the NMOS device in the first inverter turns on. This turns the PMOS device in the second inverter on. The overall delay is calculated as

$$t_{PLH} = \overbrace{0.7 \cdot 3.4k \cdot 11.57 \text{ fF}}^{\text{First inverter's delay}} + \overbrace{0.7 \cdot 6.8k \cdot 56.88 \text{ fF}}^{\text{Second inverter's delay}} \approx 300 \text{ ps}$$

Similarly, when the input goes low, the PMOS in the first inverter turns on and the NMOS in the second inverter turns on, pulling the output low

$$t_{PHL} = \overbrace{0.7 \cdot 6.8k \cdot 11.57 \text{ fF}}^{\text{First inverter's delay}} + \overbrace{0.7 \cdot 340 \cdot 56.88 \text{ fF}}^{\text{Second inverter's delay}} \approx 70 \text{ ps}$$

The simulation results are seen in Fig. 13.30. ■

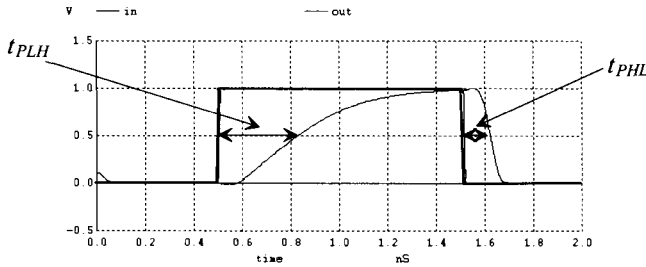


Figure 13.30 Simulating the operation of the circuit in Fig. 13.29.

Example 13.5

Estimate the delay through the circuit seen in Fig. 13.31 using 20/1 PMOS and 10/1 NMOS devices.

Since the load capacitance, 50 fF, is much larger than the output capacitance of the NAND gate, we don't include the NAND gate's capacitance contributions in the digital model seen in Fig. 13.31c. When the input goes high, the NMOS device in the inverter turns on and the PMOS device in the NAND gate turns on (causing the output to go high). The low-to-high delay time is calculated as

$$t_{PLH} = 0.7 \cdot 3.4k \cdot 4.7f + 0.7 \cdot 3.4k \cdot 50f = 130 \text{ ps}$$

The high-to-low delay time is calculated, noting the two NMOS in series when pulling the output low, as

$$t_{PHL} = 0.7 \cdot 3.4k \cdot 4.7f + 0.7 \cdot 6.8k \cdot 50f = 250 \text{ ps}$$

Figure 13.32 shows the simulation results. ■

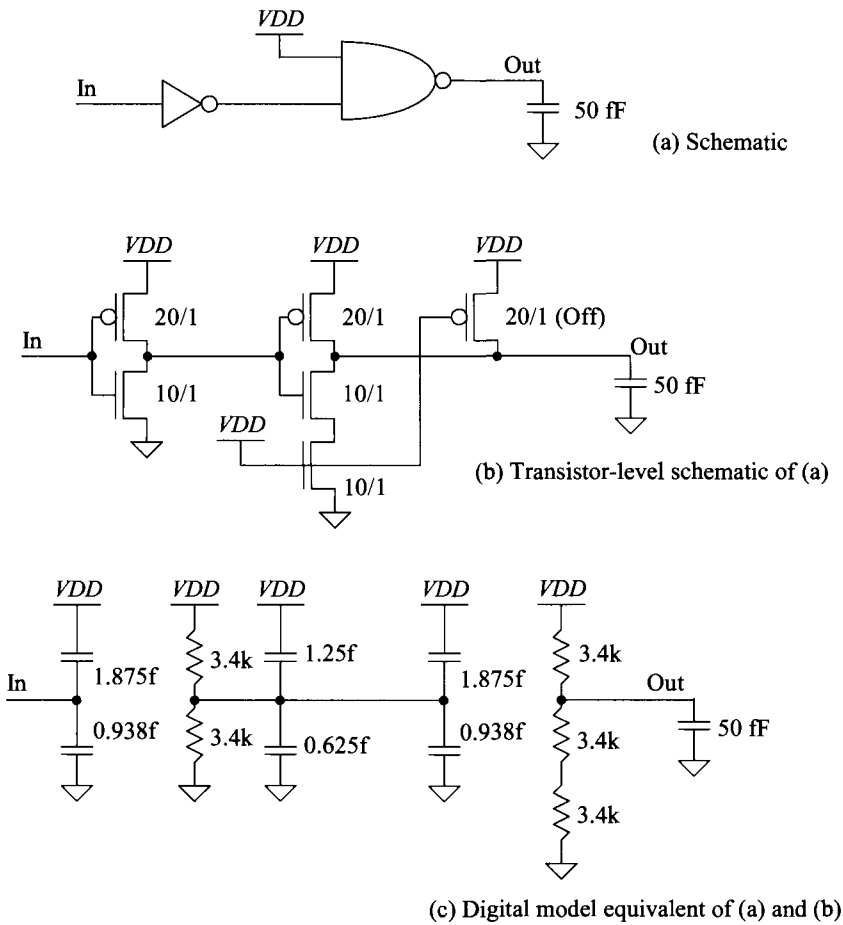


Figure 13.31 Circuit used in Ex. 13.5.

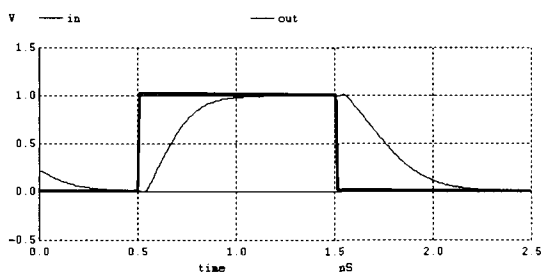


Figure 13.32 Simulating the circuit in Fig. 13.31.

Example 13.6

The circuit seen in Fig. 13.33 is the input portion of the edge triggered D-FF seen in Fig. 13.22. Estimate the delay from the D input to points A and B. Compare the estimate to SPICE simulations. Note that this delay is important because it directly determines the setup time of the FF.

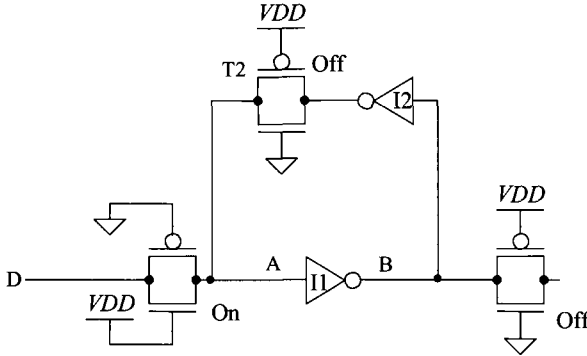


Figure 13.33 Section of a D-FF used in Ex. 13.5.

Node A in Fig. 13.33 is driven through the on TG. The resistance of the TG is $R_n || R_p$. The capacitance on node A is

$$C_A = \underbrace{\frac{C_{oxn}}{2} + \frac{C_{oxp}}{2}}_{\text{Capacitance from the on TG}} + \underbrace{\frac{3}{2} \cdot (C_{oxn} + C_{oxp})}_{\text{Input capacitance of the inverter}} = 3.75 \text{ fF}$$

The first term is the capacitance on node A due to the on TG (see Fig. 10.7). The second term is the input capacitance of the inverter. Note that we have not included the capacitive loading from the off TG. This causes the actual delays to be longer than what we calculate here. (A good exercise at this point is to estimate the capacitive loading of the TGs and include them in the calculations here.) The capacitance on node B is made up of an inverter input capacitance and an inverter output capacitance or

$$C_B = \underbrace{\frac{3}{2} \cdot (C_{oxn} + C_{oxp})}_{\text{Inverter input capacitance}} + \underbrace{(C_{oxn} + C_{oxp})}_{\text{Inverter output capacitance}} = 4.7 \text{ fF}$$

The delay from the D input to point A is estimated as

$$t_{PLHA} = t_{PHLA} = 0.7 \cdot R_n || R_p \cdot C_A = 4.5 \text{ ps}$$

The delay from point A to point B through the inverter is

$$t_{PLHB} + t_{PHLB} = 0.7 \cdot (R_n + R_p) \cdot C_B = 22.5 \text{ ps}$$

The propagation delay from D to B is then

$$t_{PLH} = t_{PHL} = 15.75 \text{ ps}$$

The simulation results are seen in Fig. 13.34. The delay to point A is approximately 30 ps, while the delay to point B is about 90 ps. It's important to reiterate the importance of simulations when determining delays. Hand calculations, as this example shows, have limitations. Hand calculations are still important, however, because they reveal the location of the dominant delays in a digital circuit. ■

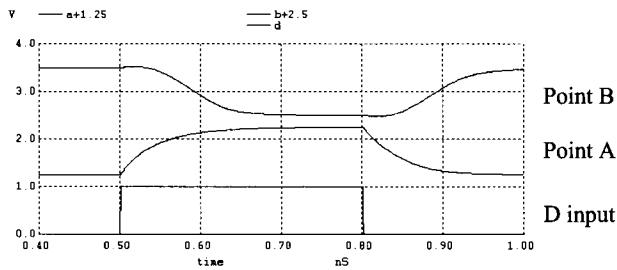


Figure 13.34 Simulating the delays through the latch seen in Fig. 13.33.

ADDITIONAL READING

- [1] M. I. Elmasry, *Digital MOS Integrated Circuits II*, IEEE Press, 1992. ISBN 0-87942-275-0, IEEE order number: PC0269-1.
- [2] J. P. Uyemura, *Circuit Design for Digital CMOS VLSI*, Kluwer Academic Publishers, 1992.
- [3] M. Shoji, *CMOS Digital Circuit Technology*, Prentice-Hall, 1988. ISBN 0-13-138850-9.

PROBLEMS

Unless otherwise stated, use the 50 nm, short-channel CMOS process with the parameters seen in Table 10.2.

- 13.1** Estimate and simulate the delay through 10 TGs connected a 50 fF load capacitance.
- 13.2** Design and simulate the operation of a half adder circuit using TGs.
- 13.3** Sketch the schematic of an 8-to-1 DEMUX using NMOS PGs. Estimate the delay through the DEMUX when the output is connected to a 50 fF load capacitance.
- 13.4** Verify, using SPICE, that the circuit seen in Fig. 13.12 operates as an XOR gate.
- 13.5** Simulate the operation of an SR latch made with NAND gates. Show all four possible input logic combinations.
- 13.6** Simulate the operation of the arbiter seen in Fig. 13.15. Show how two inputs arriving at nearly the same time results in only one output going high.